

Multimodal Driver Fatigue Detection

Literature review of 50 journal papers, five research gaps, project objectives, and a guide to the domain and its key concepts

Project: real-time driver fatigue detection fusing webcam facial cues with driving-simulator telemetry

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How to use this document

Every paper in this review is a peer-reviewed journal article checked against its Crossref record. The five research gaps are drawn from limitations the papers themselves report, and each objective is mapped to one gap with a measurable success criterion. Section 7 is a study guide for explaining the domain, the formulas and the system.

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1. Introduction and scope

The project is a real-time driver fatigue monitor that fuses two non-intrusive data streams. A webcam measures eye closure, blinking, yawning and head pitch from MediaPipe face landmarks. A browser driving simulator measures lane keeping, steering behaviour and reaction time to wind gusts. Both streams are summarised over a 30-second sliding window into a 19-feature vector that a LightGBM classifier scores once per second. The score is smoothed and mapped to four alert levels.

This document reviews the journal literature behind each of those design choices, identifies five research gaps that the project responds to, frames realistic objectives against them, and ends with a study guide for explaining the domain and the system with confidence.

Review method

- **Inclusion:** peer-reviewed journal articles only. Conference papers, preprints, theses and book chapters were excluded, even well-known ones such as the original Eye Aspect Ratio workshop paper.
- **Verification:** every paper was checked against its Crossref record (type *journal-article*). Authors, title, journal, volume, issue and pages come from that record, and the DOI resolves. Summaries are based on the abstracts.
- **Themes:** papers were grouped into five themes that mirror the system: vision-based detection, driving-behaviour measures, physiological measures and ground truth, multimodal fusion, and machine learning, datasets, real-time deployment and alerting.
- **Citation counts** are from Semantic Scholar at the time of writing and are a rough guide to influence only.

Theme	Papers	What it informs in the project
Vision-based detection	8	EAR, PERCLOS, blink and yawn features; robustness to lighting and pose
Driving-behaviour measures	9	Lane drift, steering reversals, reaction delay; simulator validity
Physiological measures, ground truth and broad reviews	10	Why contact-free sensing; how ground-truth labels are obtained; problem size
Multimodal fusion	10	Feature-level camera plus vehicle fusion; missing-modality handling
Machine learning, datasets, real-time deployment and alerting	13	LightGBM choice, evaluation protocol, real-time limits, graded alerts
Total	50	

2. The system under review

The diagram shows the data flow as implemented in the repository. Each later section links literature back to a box in this picture.

Two non-intrusive streams are fused every second into one fatigue probability

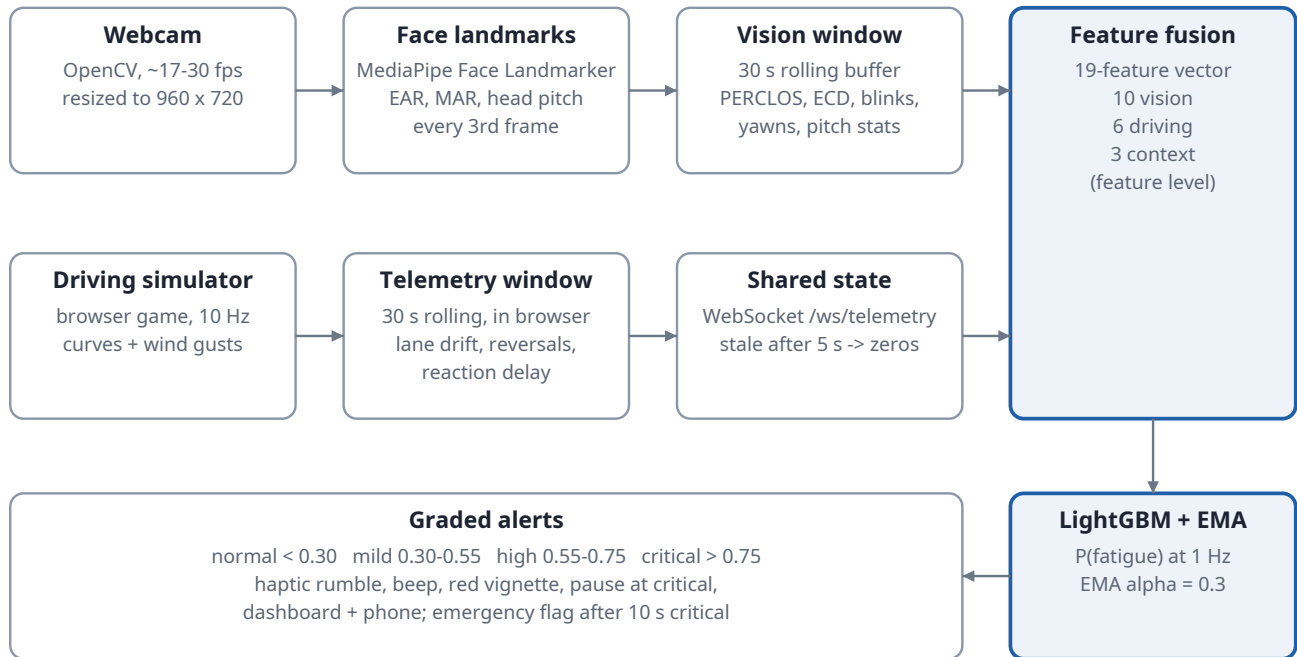


Figure 1. Data flow of the project. Vision features are computed in `backend/vision`, telemetry features in `simulator/index.html`, fusion and scoring in `backend/pipeline_interface.py`.

3. Literature review by theme

Each theme opens with a short synthesis of what the papers agree on and where they fall short, then lists the papers with their method, key finding and relevance to this project. Full references are at the end.

3.1 Vision-based detection

Eye-based measures are the most consistent camera signal of drowsiness. A systematic review of blink parameters found that almost all varied with drowsiness, with blink duration and PERCLOS the most robust (Cori et al., 2019). Broader surveys of camera-based monitoring reach the same conclusion and add yawning and head movement as common cues (Ramzan et al., 2019; Albadawi et al., 2022).

The recurring weakness is robustness. CNN-based methods degrade under cab lighting changes, shadows, occlusion and head rotation, and they confuse talking with yawning and normal blinks with drowsy closures (Bai et al., 2022; Yang et al., 2021). Results come mostly from posed or laboratory datasets, and models generalise poorly across people: one eye-based system needed a per-user feedback step (Maior et al., 2020), and a camera model reached 92% only after personalisation versus 72% without it (Bakker et al., 2022). Temporal models on eye data perform well in simulators (Quddus et al., 2021).

For this project: the literature supports EAR-based blink and closure features, PERCLOS and yawn detection as the vision core, and warns that fixed thresholds, lighting and the yawn/talk confusion are the main threats. The project's hysteresis thresholds and 30 s aggregation address noise, but per-driver calibration and lighting robustness remain open.

#	Study	Method	Key finding	Relevance to this project
1	Albadawi et al. (2022) <i>Sensors</i>	Review of driver drowsiness detection systems from the last decade, grouped into four categories by information used, with the features, classifiers and datasets of each.	Compares systems by classification accuracy, sensitivity and precision. Discusses the practicality and reliability of each of the four system types and lists current challenges and future trends.	Recent overview of features, classifiers and datasets used to benchmark this project's 19-feature LightGBM fusion against other systems. Also supports the discussion of practicality and real-time constraints.
2	Bai et al. (2022) <i>IEEE Transactions on Cybernetics</i>	Facial landmarks extracted from real-time video are fed to a two-stream spatial-temporal graph convolutional network (2s-STGCN) that models first- and second-order landmark information. Evaluated on YawDD and NTHU-DDD.	Average accuracy was 93.4% on YawDD and 92.7% on the NTHU-DDD evaluation set. The method processes whole videos rather than consecutive frames to tell talking from yawning and blinking from eye closure.	Supports a landmark-based representation (as with MediaPipe's 478 landmarks) and temporal modelling over windows. Highlights lighting, occlusion and head-pose robustness as risks for the project's EAR/MAR pipeline.

#	Study	Method	Key finding	Relevance to this project
3	Bakker et al. (2022) <i>IEEE Transactions on Intelligent Transportation Systems</i>	Video-only (single modality) two-stage model: generic deep feature extraction plus a personalised sleepiness module; 13 drivers, day vs night 90 min sessions on a Swedish motorway; KSS regression and binary classification.	Personalised binary classification reached 92% accuracy (sens 91.7%, spec 92.3%) with a KSS MAE of 0.54. Without personalisation accuracy was 72%, and a PERCLOS baseline reached 68% on the same data.	Counterpoint / challenge: a single camera modality with personalisation can match fused systems on real roads, while generic PERCLOS performs poorly. This argues for personalisation and against relying on PERCLOS alone.
4	Quddus et al. (2021) <i>Accident Analysis & Prevention</i>	48x48 eye patches from an ordinary camera (no eye tracker) for 38 subjects in a driving simulator. Models eye movement with a 1-D LSTM (R-LSTM) and a convolutional LSTM (C-LSTM). Alert/drowsy labels come from EEG power spectral analysis.	R-LSTM reached about 82% accuracy and C-LSTM 95-97%, beating a recently published eye-tracking-based approach by a wide margin. The authors note that eye-based sensing balances EEG accuracy against the ease of vehicle dynamics.	Supports camera-only eye monitoring over dedicated eye trackers. Offers a learned alternative to handcrafted EAR/PERCLOS features. Its reliance on EEG labels contrasts with this project's synthetic training labels.
5	Yang et al. (2021) <i>IEEE Transactions on Multimedia</i>	3D-CNN plus bidirectional LSTM for spatiotemporal features, with a low-cost keyframe selection step (image histograms and median absolute deviation). Evaluated on YawDD and self-collected datasets.	Several facial actions and expressions deform the mouth the same way yawning does. The proposed method achieved high yawn detection rates compared with state-of-the-art methods and distinguished yawning from similar facial actions.	Challenges the project's threshold-based MAR yawn detector. A static MAR threshold may confuse talking or laughing with yawning, which argues for duration and temporal constraints on MAR.
6	Maier et al. (2020) <i>Expert Systems with Applications</i>	Low-cost webcam pipeline that computes EAR over several consecutive frames and applies blink-pattern drowsiness rules. Compares MLP, random forest and SVM on the DROZY database, using KSS and reaction time as performance indicators.	SVM performed best overall, with 94.9% average test accuracy and the best execution time. In the inter-subject analysis it issued no warnings for awake subjects and an average of 16.1 warnings for drowsy subjects (94.44% accuracy). A per-user feedback step improved results further.	Closest precedent for this project's webcam EAR pipeline and temporal windowing. Supports EAR-based blink features and per-user calibration of EAR thresholds.
7	Cori et al. (2019) <i>Sleep Medicine Reviews</i>	Narrative review of studies that use spontaneous eye-blink parameters to measure state drowsiness. Studies differ in design, blink acquisition technology and the parameters assessed. Includes obstructive sleep apnoea (OSA) populations.	Almost all blink parameters varied consistently with drowsiness, and blink duration and percentage of eye closure (PERCLOS) were the most robust. Blink parameters were associated with and predicted conventional drowsiness measures with generally fair to good accuracy.	Directly validates choosing PERCLOS and max eye-closure duration as core vision features. Supports weighting blink duration and closure over raw blink frequency.

#	Study	Method	Key finding	Relevance to this project
8	Ramzan et al. (2019) <i>IEEE Access</i>	Survey that sorts drowsiness detection into behavioural (yawning, eye closure, head movement), vehicular and physiological techniques, and reviews the supervised learning classifiers used.	Facial cues such as yawning, eye closure and head movement are widely used to infer drowsiness level. The survey compares the pros and cons of behavioural, vehicular and physiological methods and of common supervised classifiers.	Supports this project's combination of eye closure (EAR/PERCLOS), yawning (MAR) and head movement (pitch) as behavioural features, and the use of a supervised classifier.

3.2 Driving-behaviour measures

Vehicle-based measures show that sleepiness degrades lane keeping and steering, with stronger effects in monotonous, long drives (Soares et al., 2020). Lateral lane variability (SDLP) correlates best with fatigue, and steering variability carries much of the same information (Forsman et al., 2013). Reaction times to hazards lengthen after sleep deprivation, although the size of the effect depends on the scenario (Mahajan & Velaga, 2020).

Three limitations recur. First, road context confounds the signal: curves and road type create false alarms (McDonald et al., 2018), an online steering-entropy system misclassified 29% of awake periods (Li et al., 2017), and some studies use only straight segments (Forsman et al., 2013). Second, driving-only indicators plateau around 0.72 to 0.74 and improve only when combined with other information (Sandberg et al., 2011); the field has relied on simple averaged metrics (Liu et al., 2009). Third, simulators raise sleepiness and exaggerate lateral variability compared with real roads (Hallvig et al., 2013), and only about half of simulator validation studies show validity (Wynne et al., 2019).

For this project: the steering features are computed on steering error relative to the curve, which targets the road-geometry confound directly; the wind-gust probe gives a reaction-time measure; and combining these with eye features follows the evidence that driving-only measures plateau. Simulator validity is a limitation to state openly.

#	Study	Method	Key finding	Relevance to this project
9	Mahajan & Velaga (2020) <i>Accident Analysis & Prevention</i>	50 male participants drove a simulator at baseline and after one and two nights of partial sleep deprivation (4.25 h, actigraphy-verified); response time and total braking time in pedestrian-crossing and vehicle-merging hazards modelled with Weibull accelerated failure time models.	In the pedestrian event RT rose 10% after the first PSD session (no significant effect in the second); in the vehicle-merging event response time and total braking time were delayed by 44% and 17% after PSD. Age, experience, work-rest hours, KSS/SSQ and approach speed were also significant.	Supports the reaction-delay (to wind gust) feature, but shows reaction-time effects are scenario-dependent, so a single gust-response delay may be a noisy fatigue indicator.

#	Study	Method	Key finding	Relevance to this project
10	Soares et al. (2020) <i>Traffic Injury Prevention</i>	PRISMA systematic review of simulator-based drowsy/fatigued driving studies; 23 studies covering 690 participants.	Drowsiness impairs driving performance, with effects becoming more pronounced with time-on-task and in monotonous scenarios and landscapes; some studies validated technologies to detect and predict sleepiness.	Supports using a browser driving simulator and monotonous road design to induce/observe fatigue; time-on-task effects argue for windowed temporal features (30 s window) and trend tracking (EMA).
11	Wynne et al. (2019) <i>Safety Science</i>	Systematic review of 44 studies directly comparing simulated driving with on-road driving on measures such as mean speed, speed variability, lateral position, overall performance and errors.	Simulators achieved absolute or relative validity in roughly half of studies while about one third produced non-valid results; the relationship between simulator fidelity and validity was not straightforward.	Directly challenges the project's low-fidelity browser simulator: lateral-position and speed metrics may only have relative (not absolute) validity, so thresholds derived from it may not transfer to real vehicles.
12	McDonald et al. (2018) <i>Accident Analysis & Prevention</i>	72 participants in the National Advanced Driving Simulator; steering angle, pedal input, speed and acceleration fed to a Dynamic Bayesian Network with a real-time driving-context measure to detect drowsiness-related lane departures.	The algorithm had a significantly lower false positive rate than PERCLOS and non-contextual baselines, reducing false positives in highway and rural environments that are typically problematic for vehicle-based detectors.	Challenges the project's context-free features (steering corrections partly reflect road demands, e.g., wind gusts/curves) and supports temporal modelling of state transitions beyond EMA smoothing.
13	Li et al. (2017) <i>Sensors</i>	Real-road study (14.68 h of driving); approximate entropy features of steering wheel angle extracted from fixed sliding windows, linearised, compared by warping distance and classified awake/drowsy with a binary decision classifier.	The online system achieved 78.01% average accuracy, with 29.35% false detections of the awake state and 15.15% false detections of the drowsy state.	Supports steering-entropy/steering-instability features computed over fixed sliding windows (cf. the 30 s window) and shows steering-only systems produce many false alarms, motivating EMA smoothing and multimodal fusion.
14	Forsman et al. (2013) <i>Accident Analysis & Prevention</i>	87 literature drowsiness metrics evaluated in two simulated shift-work studies (29 night-shift, 12 day-shift participants; four 30-min high-fidelity simulator drives per day) using PCA, validated against PVT performance.	Two dominant dimensions emerged: steering wheel variability and lateral lane position variability, with the latter correlating most with PVT fatigue; lane variability could be derived from steering angle via a transfer function, making steering variability a cheap basis for detection at moderate fatigue.	Strongly supports the steering instability and lane drift variance features and early (moderate) detection; the road-segment restriction highlights road-geometry confounds for the project's simulator.

#	Study	Method	Key finding	Relevance to this project
15	Hallvig et al. (2013) <i>Accident Analysis & Prevention</i>	10 participants drove day and night on a real road and in a high-fidelity moving-base simulator emulating the same road; driving performance, EEG/EOG and subjective sleepiness compared.	The simulator produced higher subjective and physiological sleepiness than real driving; lateral variability was more responsive to night driving in the simulator, whereas real night driving involved a leftward lane shift and speed reduction absent in the simulator. Relative validity acceptable for many variables, but absolute levels differ.	Key evidence for the simulator-validity gap: lane drift variance (SDLP) may overstate impairment in a simulator; speed reduction and lane-position shift are compensatory real-road behaviours the project's features do not capture.
16	Sandberg et al. (2011) <i>IEEE Transactions on Intelligent Transportation Systems</i>	12 participants, 12 one-hour sessions each in the VTI moving-base simulator; literature driving-behaviour indicators optimised stochastically and combined with a sleep/wake predictor (SWP) via a neural network.	The best literature indicator (SD of yaw angle) scored 0.72 (mean of sensitivity and specificity); a generalised variability indicator reached 0.74, SWP alone 0.78, and the nonlinear combination of SWP with the variability indicator 0.83.	Shows a ceiling for single driving-variability features (lane/steering variance), supporting fusion with other modalities (the project fuses with eye/face features) and suggesting a circadian/time-awake prior could add value.
17	Liu et al. (2009) <i>Journal of Safety Research</i>	Narrative review assessing whether vehicle measures (e.g., standard deviation of lane position, steering) can reliably predict drowsiness in real time.	Drowsiness seriously degrades driving performance in controlled experiments, but most studies used simple performance functions such as SD of lane position and reported averages across drivers and time; a successful countermeasure will likely need multiple criteria and multiple measures.	Seminal justification for vehicle-based features (SDLP/lane drift variance, steering) and for multi-measure fusion; flags the individual-differences problem that a single LightGBM trained on synthetic data does not address. (Older seminal paper, pre-2010.)

3.3 Physiological measures, ground truth and broad reviews

Physiological signals give the most direct view of the brain and body. EEG theta and alpha changes track the shift from workload to fatigue to drowsiness, with detection around 90% accuracy, but offline (Borghini et al., 2014). EEG alpha spindles separated fatigued drivers in real traffic better than band power (Simon et al., 2011), and heart-rate variability flagged 12 of 13 sleep-onset episodes before EEG-scored sleep, at 1.7 false alarms per hour (Fujiwara et al., 2019). All need body-contact electrodes.

Ground truth is the deeper problem. The Karolinska Sleepiness Scale is the standard label (Åkerstedt & Gillberg, 1990), it rises sensitively with sleep loss and night driving, but it shifts by one to two points with context and relates to physiology in a strongly curvilinear way (Åkerstedt et al., 2014). Its correlation with objective lapses is moderate ($r = 0.56$) (Kaida et al., 2006). Trained observer ratings of face video are a reliable alternative (Wierwille & Ellsworth, 1994). The problem is real: sleepiness at the wheel more than doubles crash risk (pooled odds ratio 2.51) (Bioulac et al., 2017).

Broad reviews compare the measure families and agree that each has trade-offs, that promising results come mostly from constrained environments, and that hybrid, non-intrusive systems are the way forward (Sahayadhas et al., 2012; Sikander & Anwar, 2019). **For this project:** this justifies contact-free sensing and shows that any real training data will need KSS or observer-rating labels,

with their noise acknowledged.

#	Study	Method	Key finding	Relevance to this project
18	Fujiwara et al. (2019) <i>IEEE Transactions on Biomedical Engineering</i>	Driving-simulator study of 34 participants in which eight HRV features from ECG RR intervals were monitored with multivariate statistical process control (anomaly detection). Detections were validated against sleep onsets scored from EEG by a sleep specialist.	The HRV algorithm detected drowsiness in 12 of 13 pre-N1 episodes before EEG-scored sleep onset, with a false-positive rate of 1.7 per hour.	A clear example of validating against an EEG gold standard. It reports event-level detection with false alarms per hour, a metric the project could adopt alongside accuracy to judge alert nuisance. It contrasts contact-based ECG sensing with the project's contact-free approach.
19	Sikander & Anwar (2019) <i>IEEE Transactions on Intelligent Transportation Systems</i>	State-of-the-art review that sorts fatigue detection methods into five groups: subjective reporting, driver biological features, driver physical (camera/facial) features, vehicular features and hybrid features.	Several published methods report promising results, but mostly in constrained environments. Much progress is still needed, including on commercial in-vehicle and third-party devices. The review compares approaches and identifies areas open for improvement.	Gives a taxonomy that places this project in the 'hybrid features' category (physical/facial features plus vehicular telemetry). Supports the motivation for fusing EAR/PERCLOS/MAR/head pose with lane and steering features.
20	Bioulac et al. (2017) <i>Sleep</i>	PROSPERO-registered systematic review and random-effects meta-analysis of adjusted risk estimates linking self-reported sleepiness at the wheel to motor vehicle accidents, using studies from Medline, Scopus and Web of Science.	Pooling 10 cross-sectional studies (51,520 participants), 6 case-control studies (4,904) and 1 cohort study (13,674), the review found that sleepiness at the wheel was linked to higher motor vehicle accident risk (pooled OR 2.51, 95% CI 1.87-3.39). Heterogeneity between studies was high ($Q = 93.21$; $I^2 = 83\%$).	Supplies the problem-statement statistic: drowsiness at the wheel roughly 2.5-folds the odds of a crash. It also shows that even large-scale epidemiology depends on subjective sleepiness reports, which supports the case for objective, continuous monitoring.
21	Borghini et al. (2014) <i>Neuroscience & Biobehavioral Reviews</i>	Review of studies that measured EEG, EOG and heart rate in aircraft pilots and car drivers during operation, relating neurophysiological changes to mental workload, mental fatigue and drowsiness.	The literature shows a consistent sequence: high workload raises theta and lowers alpha, the shift to mental fatigue raises theta, delta and alpha power, and drowsiness brings a higher blink rate and lower heart rate. Detection of these states reaches about 90% accuracy, but only offline, not online.	Establishes EEG/EOG/HR as the reference physiology that non-intrusive systems are judged against. The blink-rate finding supports camera-based eye metrics as a contact-free proxy, and the 'offline, not online' gap motivates the project's real-time pipeline.

#	Study	Method	Key finding	Relevance to this project
22	Åkerstedt et al. (2014) <i>Journal of Sleep Research</i>	Narrative review of evidence on the sensitivity, reliability and validity of the 9-point Karolinska Sleepiness Scale against sleep loss, night work, driving performance and EEG/EOG indicators of sleepiness.	KSS rises sensitively under acute and repeated partial sleep deprivation and night work, including night driving (effect sizes 1.5-3). Its relation to driving performance and EEG/EOG is highly significant, strongly curvilinear and consistent across individuals, with KSS > 6 linked in particular to impaired driving and EEG sleep intrusions. Context matters: physical activity, social interaction and light lower ratings by 1-2 units, while monotonous time-on-task raises them by 1-2 units.	The strongest single justification for KSS-based labels and for placing the alert-level cut-off around KSS > 6 when mapping P(fatigue) to the 4 alert levels. The context effects warn that simulator monotony may inflate fatigue, which matters for how synthetic sessions are parameterised.
23	Sahayadhas et al. (2012) <i>Sensors</i>	Review of the sensors used for vehicle-based, behavioural (camera-based facial and eye) and physiological drowsiness measures, plus how drowsiness has been induced in experiments.	Each of the three measure types has its own advantages and limitations. The authors conclude that a hybrid system combining non-intrusive physiological measures with other measures would determine driver drowsiness level most accurately.	Directly supports this project's multimodal fusion of behavioural (camera) and vehicle-based (lane and steering telemetry) measures. Also points to a gap: physiological signals are not included in this project.
24	Simon et al. (2011) <i>Clinical Neurophysiology</i>	The authors built an algorithm to detect EEG alpha spindles (short narrowband alpha bursts) and tested it on simulated data. They then applied it to EEG recorded in real-traffic driving, comparing the first and last 20 minutes of drivers who stopped because of heavy fatigue, and benchmarked it against traditional alpha-band power.	Several alpha-spindle parameters rose significantly from the first to the last driving section, with larger effect sizes than alpha-band power, the only traditional EEG band that changed. Only spindle parameters distinguished drivers who aborted the drive from those who completed it, so spindles improved both sensitivity and specificity over band power.	Shows that even the 'gold standard' EEG depends on feature choice, and that its own ground truth is a proxy (time-on-task, drop-out). This supports treating labels as uncertain and including time-on-task in the synthetic fatigue model. It is also one of the few EEG validations done in real traffic rather than a simulator.
25	Kaida et al. (2006) <i>Clinical Neurophysiology</i>	Sixteen healthy women aged 33-43 completed 8 sessions a day for 3 consecutive days, each with the psychomotor vigilance task (PVT), the Karolinska Drowsiness Test (EEG alpha/theta power), the alpha attenuation test and the KSS. KSS was then related to these measures.	Median reaction time, number of lapses, alpha and theta power density and alpha attenuation coefficients all rose significantly as KSS increased and correlated with KSS (mean $r = 0.56$ for lapses). The authors conclude that KSS ratings may be a useful proxy for EEG or behavioural indicators of sleepiness.	The main validation evidence for using the KSS as a ground-truth proxy for EEG. It supports anchoring the synthetic data's latent fatigue variable to a KSS-like scale. The finding that reaction-time lapses track KSS also supports the project's reaction-delay telemetry feature.

#	Study	Method	Key finding	Relevance to this project
26	Wierwille & Ellsworth (1994) <i>Accident Analysis & Prevention</i>	Trained observer-raters scored drivers' drowsiness level from videotaped facial segments in two sessions a week apart, to measure test-retest, inter-rater and intra-rater reliability and consistency. A follow-up experiment compared the ratings with other drowsiness indicators.	Observer ratings of drowsiness from face video were reliable and consistent across the reliability measures, and a follow-up experiment showed that the ratings covaried with other known indicators of drowsiness.	Foundational for observer rating of drowsiness (ORD), the video-based labelling approach closest to a camera-only system. It justifies having raters label webcam recordings of real users to validate the classifier trained on synthetic data. The facial cues raters judge (eye closure, yawning, head nodding) correspond to the project's EAR/PERCLOS/MAR/head-pitch features.
27	Åkerstedt & Gillberg (1990) <i>International Journal of Neuroscience</i>	Eight participants were kept awake and active overnight in a time-isolated sleep lab. The study recorded continuous ambulatory EEG/EOG, sleepiness ratings every two hours and short eyes-open/eyes-closed EEG tests, then applied EEG spectral analysis and visual scoring of slow eye movements (SEM).	During open-eyed activity, intrusions of SEM and of alpha/theta power clearly separated high from low subjective sleepiness, and mean within-person correlations between subjective and objective sleepiness were very high. However, the relation was curvilinear: physiological signs appeared reliably only once ratings fell between 'sleepy' and 'extremely sleepy - fighting sleep'.	This is the paper conventionally cited as the origin of the 9-point Karolinska Sleepiness Scale (the abstract itself says only 'sleepiness ratings'). The curvilinear, threshold-like link between self-rated sleepiness and EEG/EOG signs supports using a non-linear classifier (LightGBM). It also suggests that the synthetic data generator should make ocular and physiological effects appear mainly at high fatigue levels, not linearly.

3.4 Multimodal fusion

Fusion studies generally beat their best single modality. PERCLOS fused with a vehicle heading-error measure reached $f = 0.96$ against 0.86 for PERCLOS alone (Daza et al., 2014); eye, steering and lane fusion reached 94.6% against 90.7% for the best single input (Samiee et al., 2014); and a two-stage camera and vehicle system was significantly more reliable than any single sensor (Cheng et al., 2012). In a 23-variable model, PERCLOS, pupil diameter, SDLP and steering reversals were the most important variables (Wang & Xu, 2016), closely matching this project's feature set.

Many fusion systems still depend on intrusive sensors: physiological signals with face video (Peng et al., 2024), or EEG, ECG and EOG with driving data (Martensson et al., 2019; Jacobé de Naurois et al., 2019). Encouragingly, dropping the physiological sensors and keeping only vehicle and behavioural signals retained most of the accuracy (78.7% to 89.8% versus 82.4% to 95.4%) (Gwak et al., 2020), and a wearable added little in another study (Schwarz et al., 2023).

Two weaknesses matter most here. Generalisation to new drivers is poor: sensitivity fell from 86.5% to 41.4% on unseen drivers (Martensson et al., 2019). And missing modalities are rarely handled: glasses-wearers were excluded (Daza et al., 2014), segments with missing data discarded (Schwarz et al., 2023), and feature-level fusion is noted to be more accurate but unable to survive signal loss, unlike decision-level fusion (Samiee et al., 2014). Gated modality selection is one recent answer (Aravinth et al., 2025). **For this project:** the evidence backs non-contact camera plus

vehicle fusion, and flags its single 19-feature vector as fragile when the face is lost.

#	Study	Method	Key finding	Relevance to this project
28	Aravinth et al. (2025) <i>Scientific Reports</i>	EEG, ECG and facial video. Domain-adversarial feature extraction, a transformer with adaptive cross-modal attention (intermediate fusion), a Gated Modality Selection Network that scores sensor quality and turns modalities on or off, and online per-driver fine-tuning. Evaluated on public datasets (SEED-VIG and a PhysioNet driving-fatigue set, per the full text).	Cross-modal fusion improved accuracy by 5-8%. Gated modality selection kept the performance drop under 5% with noisy or missing inputs, overall accuracy was 85-90% on target domains under 20% sensor dropout, and online personalisation added 5-7% within 2 h at under 50 ms latency.	Second missing-modality paper: gating modalities by quality is a direct template for down-weighting the camera features when the face is not detected. Online personalisation addresses this project's generic synthetic-data model.
29	Peng et al. (2024) <i>IEEE Transactions on Intelligent Transportation Systems</i>	End-to-end neural multi-source fusion of physiological signals and facial video, with automatic feature extraction instead of hand-picked features; 21 participants in simulated driving.	93.15% accuracy (specificity 94.04%, sensitivity 91.71%) within a 3 s window, outperforming existing methods in accuracy and real-time performance. The authors argue that hand-picked features need long windows that limit accuracy and responsiveness.	Challenges this project's hand-crafted features over a 30 s window, arguing that learned features over short windows are faster and more accurate. Useful for discussing the latency vs interpretability trade-off.
30	Schwarz et al. (2023) <i>Traffic Injury Prevention</i>	Production-type camera driver-monitoring system + vehicle data from the NADS motion-base simulator + HRV from an Empatica E4 wristband; 40 participants awake for 16+ h drove up to 3 h; nine binary random forests over different data-source and ground-truth combinations.	Model accuracy ranged from 0.77 to 0.92. Adding physiological measures added only a small amount of accuracy, the best model used a reduced dataset with missing-data segments removed, and the most timely model flagged drowsiness 6.7 min before a drowsy lane departure.	Supports camera + vehicle as the core pair (the wearable added little) and shows missing data is a practical problem in multi-sensor fusion. Its lead-time metric suggests one way to evaluate this project's alert levels.
31	Gwak et al. (2020) <i>Applied Sciences</i>	Three-way hybrid of vehicle-based (driving performance), behavioural (driver-monitoring camera) and physiological (EEG, ECG) indices extracted over 10 s windows in a driving simulator, classified with ensemble learners (random forest and others).	The hybrid ensemble reached 82.4% for alert vs slightly drowsy and 95.4% for alert vs moderately drowsy. Excluding physiological indicators (vehicle + behavioural only), random forest still reached 78.7% and 89.8% respectively.	Direct evidence that a non-contact camera + vehicle combination (the configuration of this project) keeps most of the accuracy of a full hybrid, especially for moderate drowsiness. Early (slight) drowsiness remains harder, which bears on this project's lowest alert levels.

#	Study	Method	Key finding	Relevance to this project
32	Jacobé de Naurois et al. (2019) <i>Accident Analysis & Prevention</i>	21 participants, 110 min monotonous simulator drive. Physiological (HR/HRV, respiration), behavioural (head and eyelid movements: blink duration, frequency, PERCLOS) and driving data (TLC, speed, steering angle, lane position), in various combinations, fed to ANNs for minute-by-minute detection and prediction against Trained Observer Ratings.	The best detection and prediction came from behavioural indicators plus additional information (driving time, participant information): drowsiness level was detected with MSE 0.22, and the time to reach moderate drowsiness was predicted with MSE 4.18 min.	Supports comparing modality combinations and shows camera-based behavioural cues carry most of the signal. It suggests adding contextual features (time on task) and a prediction horizon beyond this project's detection-only alert levels.
33	Martensson et al. (2019) <i>IEEE Transactions on Intelligent Transportation Systems</i>	86 drivers on real roads in real traffic, alert and sleep-deprived; EEG, ECG, EOG and driving performance (steering, lane position) features selected by SFFS and tested with five classifiers against KSS labels (≤ 6 vs ≥ 8).	Random forest was most robust with 94.1% accuracy (sensitivity 86.5%, specificity 95.7%), but sensitivity dropped to 41.4% on drivers not in the training set; adding a biomathematical sleepiness model raised participant-independent sensitivity to 66.2%.	Supports tree-ensemble classifiers (cf. LightGBM) on fused driving + driver-state features, but warns that cross-driver generalisation is poor, a key risk for a model trained on synthetic data.
34	Wang & Xu (2016) <i>Accident Analysis & Prevention</i>	Participants drove a fixed road course in a high-fidelity motion-based simulator after an 8-h night shift; 19 driving-behaviour and 4 eye-feature variables vs KSS ratings, modelled with multilevel ordered logit, ordered logit and ANN.	The multilevel ordered logit model (accounting for individual differences) had the highest detection accuracy; PERCLOS, average pupil diameter, SDLP and steering wheel reversals were the most important of the 23 variables.	Closest analogue to this project: fuses PERCLOS with SDLP (lane drift variance) and steering reversals. Validates the feature choice but challenges a population-level model without per-driver adaptation.
35	Daza et al. (2014) <i>Sensors</i>	Custom vision system (PERCLOS) + ADAS driving-performance indicators (lateral position, time to lane crossing, steering, heading error), optimised stochastically and fused at the indicator level with a feed-forward neural network; 9 professional drivers in a third-generation simulator; supervised-KSS ground truth.	From the full text (open access): PERCLOS was the best single indicator ($f = 0.86$; spec 0.90, sens 0.82), the best vehicle indicator was optimised heading-error MSE ($f = 0.80$), and the best fusion PERCLOS + heading-error MSE reached $f = 0.96$ (spec 0.98, sens 0.94). Steering-wheel indicators alone were near chance.	Closest analogue to this project: feature-level fusion of camera eye closure with simulator lane/steering features beats either alone. Its glasses failure case directly motivates a fallback for when the face/eyes are not visible.

#	Study	Method	Key finding	Relevance to this project
36	Samiee et al. (2014) <i>Sensors</i>	Three non-intrusive detectors (camera eye-status analysis, steering angle, lateral position) each modelled with an ANN, then fused at decision level with accuracy-proportional weights and availability flags so that missing inputs are reweighted; 12 sleep-controlled volunteers in a simulator, 20 s windows.	From the full text (open access): the single methods reached 90.7% (eye), 87.22% (steering) and 85.37% (lateral position), and fusion reached 94.63%. With one signal lost, accuracy fell by at most 6.85% (worst case 87.78% when the eye signal was lost). The accuracy gain from fusion was at most 3.89%, so its main benefit is robustness.	Key missing-modality paper. It directly challenges a single feature-level vector: its decision-level design keeps working when the camera fails (87.78%). This supports adding fallback or per-modality models for frames where the face is not visible. Its 20 s window is similar to this project's 30 s window.
37	Cheng et al. (2012) <i>Human Factors and Ergonomics in Manufacturing & Service Industries</i>	Camera eye measures (PERCLOS, maximum closure duration) + steering (percentage of non-steering) + vehicle trajectory, combined in a two-stage framework: Fisher's linear discriminant for feature-level fusion, then Dempster-Shafer evidence theory for decision-level fusion.	The two-stage fusion system reached 90.7% recognition accuracy, and its reliability and accuracy were significantly higher than those of any single sensor.	Strong direct support: the same camera (PERCLOS, closure duration) + steering + lane-trajectory feature set, showing fusion beats single sensors. It also shows a hybrid feature-plus-decision design as an alternative to this project's purely feature-level fusion.

3.5 Machine learning, datasets, real-time deployment and alerting

Tree ensembles are well established for this tabular problem. A random forest on steering data beat PERCLOS in accuracy and AUC (McDonald et al., 2014), gradient boosting on simulator lane and steering data detected sleep-deprived driving with $77 \pm 9\%$ accuracy (van der Wall et al., 2021), and a personalised random forest with road information raised accuracy from 81.3% to 91.6% (Yi et al., 2019). Deep models such as CNN-LSTM reach 96% on three-class vehicle data (Arefnezhad et al., 2020) but need more data. No reputable journal paper applying LightGBM to driver drowsiness was found, so the project's choice rests on the strength of tree ensembles on tabular features rather than direct precedent.

Data and evaluation are the field's weak point. A recent review lists late detection, intrusiveness and subject diversity as the main setbacks, notes that most models are validated on subjective KSS labels, and calls for on-road data and consistent protocols (Perkins et al., 2023). Real-time systems exist on smartphones (93.3% with a three-stage verification) (Dasgupta et al., 2019), as threshold rules on eye and mouth features (about 92%) (Deng & Wu, 2019), and on an embedded GPU (96.6% at about 14 fps) (Florez et al., 2024); near-perfect scores on single datasets suggest limited generalisation testing, and latency is rarely reported.

Alerting research shows that how a warning is delivered matters as much as detection. Multistage alerts that signal rising urgency reduced drowsy lane departures more than single-stage alerts (Gaspar et al., 2017), and real-time drowsiness feedback improved alertness in field use (Aidman et al., 2015). Tactile warnings improve alertness and response (Meng & Spence, 2015), but vibration can annoy, and timing and location code meaning better than intensity (Petermeijer et al., 2016). False alarms create a cry-wolf effect that lowers compliance (Naujoks et al., 2016).

For this project: LightGBM fits the evidence for tree ensembles on tabular features; the four graded levels, multimodal haptic, audio and visual output, gentle mild level and 10 s persistence rule follow the alerting literature. The missing pieces are real data, subject-independent evaluation and measured latency.

#	Study	Method	Key finding	Relevance to this project
38	Florez et al. (2024) <i>Sensors</i>	CNN (DD-AI) for eye-state detection plus Mouth Aspect Ratio for yawning, deployed on an NVIDIA Jetson Nano with a near-infrared camera; trained/tested on NITYMED and compared with InceptionV3, VGG16, ResNet50V2.	DD-AI reached 99.88% on NITYMED test data, and 96.55% accuracy at about 14 fps on average in real-environment hardware tests.	Embedded benchmark: even an edge GPU reaches only ~14 fps with a CNN, which supports the project's lightweight landmark features + LightGBM on a laptop CPU. Uses MAR for yawns like the project.
39	Perkins et al. (2023) <i>IEEE Transactions on Intelligent Vehicles</i>	Survey of physiological, behavioural, vehicle-based and hybrid drowsiness detection, identifying barriers to on-road implementation.	Hybrid methods are emerging as superior; key setbacks are late detection, intrusiveness and subject diversity; behavioural monitoring is least robust (lighting, privacy); most models are validated on subjective KSS labels.	Gives the research-gap framing: label quality, on-road data scarcity and subject diversity. This supports the project's hybrid (camera + telemetry) design and also shows why synthetic labels and simulator data are a limitation.
40	van der Wall et al. (2021) <i>Traffic Injury Prevention</i>	Gradient boosting model on simulator driving features (lateral position, speed, steering speed) from 24 subjects tested sleep-deprived and well-rested; 5-fold cross-validation; probability scores used to compare against alcohol/alprazolam conditions.	The model detected sleep-deprived driving with $77 \pm 9\%$ accuracy; mean probability scores were 0.79 for alprazolam+alcohol, 0.63 for alcohol, and 0.27-0.30 for placebo, tracking expected relative drowsiness.	Closest precedent for the project's design: a gradient-boosting classifier on simulator lane/steering features that outputs a probability graded with impairment, which supports mapping P(fatigue) onto alert levels. The modest accuracy argues for adding camera features.
41	Arefnezhad et al. (2020) <i>Expert Systems with Applications</i>	CNN combined with RNN (LSTM/GRU) on five vehicle-based signals (lateral deviation, lateral acceleration, yaw rate, steering angle and velocity) to classify three drowsiness levels; 44 fixed-base simulator sessions.	The deep networks outperformed SVM and k-NN; CNN-LSTM reached the highest accuracy (96.0%) for 3-class drowsiness classification from vehicle data alone.	Challenges the LightGBM choice: temporal deep models beat classical classifiers on raw telemetry. Also supports multi-level (graded) drowsiness output and lane/steering features. The project's hand-crafted 1 Hz window features act as a lightweight stand-in for learned temporal features.

#	Study	Method	Key finding	Relevance to this project
42	Dasgupta et al. (2019) <i>IEEE Transactions on Intelligent Transportation Systems</i>	Android smartphone system with three-stage verification: PERCLOS from the front camera (NIR lighting at night), then voiced/unvoiced speech ratio, then a touch response before sounding an alarm; logs events with GPS and sends SMS to a control room and passengers.	Three-stage verification gave 93.33% drowsiness classification versus 86.66% for a single stage, running on a commodity smartphone.	Precedent for running on commodity hardware and for escalating, staged confirmation before alarms, plus remote notification (SMS), analogous to the project's phone vibration and emergency-contact trigger. Multi-stage confirmation reduces false alarms.
43	Deng & Wu (2019) <i>IEEE Access</i>	DriCare: improved face-tracking algorithm and 68-landmark facial regions to measure blinking, yawning and eye-closure duration from video, combined into a fatigue warning.	DriCare achieved around 92% accuracy using combined eye and mouth features, without body-worn devices.	Highly cited baseline for the project's camera pipeline (blink, eye-closure duration, yawn), showing landmark-based features run in real time on ordinary hardware; the project extends it with a learned fusion and telemetry.
44	Yi et al. (2019) <i>Transportation Research Part C: Emerging Technologies</i>	Personalised driving-state recognition (normal, drowsy, aggressive) from non-intrusive vehicle measurements on real-world driving data; feature-distribution and Spearman-distance analysis to decide when to personalise; five classifier families compared.	Random forest with Bayesian parameter optimisation performed best; the personalised system with road information raised overall accuracy from 81.3% to 91.6% compared with conventional non-personalised approaches.	Second independent source showing personalisation gives about +10 points; also finds random forest best among classifier families on vehicle features, which supports tree ensembles. Motivates driver-specific calibration in the project.
45	Gaspar et al. (2017) <i>Traffic Injury Prevention</i>	72 drivers in the NADS full-motion simulator on 45-min night drives (late night and early morning); 48 received one of 6 countermeasures varying in type (discrete vs staged) and modality, triggered by a steering-based drowsiness algorithm; 24 baseline; outcomes: lane departures and SDLP.	Countermeasures reduced drowsy lane departures and lane-position variability versus no countermeasure; multistage alerts that signal increasing urgency were more effective than single-stage discrete alerts, especially in early-morning drives; even a simple auditory-visual coffee-cup icon helped.	Strongest support for the project's graded 4-level alert scheme (normal/mild/high/critical) and multimodal warnings. The stated limitation supports escalating to a simulation pause/rest prompt at the critical level.

#	Study	Method	Key finding	Relevance to this project
46	Naujoks et al. (2016) <i>Accident Analysis & Prevention</i>	Driving simulator study (N = 80) with a 100% reliable advisory warning system, a 60% reliable system with false alarms (no obvious reason), or one with unnecessary alarms (plausible reason), plus a no-assistance baseline; warnings visual-only or visual-auditory.	False (but not unnecessary) alarms reduced compliance and system effectiveness, a cry-wolf effect seen only with visual-auditory warnings, and lowered ratings of the system; nonetheless, all assisted drives reduced safety-critical events and gave earlier braking than unassisted driving.	Warns that a synthetic-trained model with false positives will erode trust, mainly for loud multimodal alerts. Supports keeping mild-level alerts low-urgency (visual) and reserving rumble/beep/emergency escalation for high-confidence critical states.
47	Petermeijer et al. (2016) <i>IEEE Transactions on Intelligent Transportation Systems</i>	Literature survey of the physiology and psychophysics of vibrotactile sensation and the use of vibrotactile displays for take-over requests in automated driving.	Location and timing are better than frequency and amplitude for coding vibrotactile information; vibrations work as simple warnings but can cause annoyance, and closely spaced signals may not be understood; multimodal feedback is recommended.	Design guidance for the haptic channel: code alert level by timing/pulse pattern rather than amplitude alone, pair rumble with audio/visual (as the project does), and limit annoyance from repeated rumble.
48	Aidman et al. (2015) <i>Accident Analysis & Prevention</i>	15 Army Reserve drivers monitored by an infrared-oculography Optalert system for 4-8 weeks of regular driving; counterbalanced feedback vs no-feedback of blink-velocity-derived Johns Drowsiness Scale scores (419 feedback and 385 no-feedback periods).	Real-time drowsiness feedback reduced drowsiness scores and improved alertness and self-rated driving performance; the number and duration of drowsiness peaks per trip declined, consistent with a genuine feedback mechanism, though the effect was small and varied across the 24-h cycle.	Real-world evidence that continuous graded drowsiness feedback (not only alarms) helps, which supports showing the smoothed P(fatigue) level to the driver. The small effect size cautions against overclaiming alert benefits.
49	Meng & Spence (2015) <i>Accident Analysis & Prevention</i>	Narrative review comparing tactile with traditional visual and auditory in-vehicle warnings, including directional, dynamic and meaningful (iconic) tactile signals.	Basic tactile warnings promote driver alertness and improve forward collision avoidance compared with no warning; directional and dynamic tactile cues (varying intensity/location) can further speed responses and convey meaning.	Supports haptic rumble as an alerting channel and suggests varying vibration intensity/pattern by alert level, which matches the project's graded gamepad rumble.
50	McDonald et al. (2014) <i>Human Factors: The Journal of the Human Factors and Ergonomics Society</i>	Random forest classifier on steering-wheel-angle data with an ensemble definition of drowsiness; 72 participants in the National Advanced Driving Simulator; compared against PERCLOS.	The random forest steering algorithm achieved higher classification accuracy and AUC than PERCLOS with comparable positive predictive value, and its link to lane departures means it can be paired with mitigation systems.	Supports using tree ensembles on driving telemetry (steering/lane) and shows that driving-based features can beat PERCLOS alone, which supports the project fusing simulator telemetry with eye features in a tree-based model (LightGBM).

4. Comparing the approaches

Across the five themes the literature agrees on three points: each family of measures has a characteristic failure mode, fusion of complementary measures usually beats the best single one, and almost all evidence comes from simulators with small samples and subjective labels. The table summarises the trade-offs.

Measure family	Examples	Strengths	Weaknesses	Used here
Subjective	KSS (1 to 9), Stanford Sleepiness Scale	Cheap, validated, the usual label source	Self-report, interrupts the task, sparse in time	No (proposed as labels, Objective 4)
Physiological	EEG alpha/theta power, heart-rate variability, EOG	Direct link to brain and body state, early signs	Intrusive electrodes, motion artefacts, mostly offline	No (contact-free by design)
Behavioural (vision)	PERCLOS, blink rate and duration, yawns, head nods	Contact-free, cheap webcam, intuitive	Lighting, glasses, head pose; yawns and blinks have other causes	Yes: EAR, PERCLOS, ECD, MAR, pitch
Vehicle-based	SDLP, steering reversals and entropy, reaction time	No sensors on the driver, continuous	Confounded by road geometry and style; signs appear late	Yes: 6 simulator features
Hybrid / fused	Any combination of the above	Covers single-modality blind spots	More data needed; synchronisation; missing-modality handling	Yes: feature-level fusion

5. Research gaps

Five gaps recur across the reviewed papers. For each one: the evidence, how this project already responds, and what remains open. The open part is what the objectives in Section 6 target.

Gap	Project response so far	Still open
Gap 1. Camera sensing is fragile, and fusion systems rarely cope with a missing modality	The vision pipeline now reports camera and face status, and the telemetry stream falls back to zeros after 5 s. In testing the face was detected in only 0 to 10% of frames under backlighting, so the gap is real for this system.	Graceful degradation to a driving-only score when the face is lost; per-driver calibration of EAR and MAR thresholds; measured robustness across lighting conditions.
Gap 2. Vehicle-based measures are confounded by road context and plateau on their own	Steering instability, reversals and corrections are computed on steering error relative to the curve, a wind-gust probe measures reaction delay, and driving features are fused with ocular features rather than used alone.	Validation of the curve-normalised features against human drivers, not only autopilot profiles; testing on varied road geometry.
Gap 3. Fusion research relies on intrusive sensors and offline, simulator-only evaluation	A low-cost, contact-free loop: webcam plus simulator, feature-level fusion of exactly the variables the literature ranks highest (PERCLOS, SDLP, steering reversals) (Wang & Xu, 2016), scored live at 1 Hz on a laptop CPU.	Measured end-to-end latency (rarely reported; one embedded system managed about 14 fps (Florez et al., 2024)) and an ablation showing whether fusion beats each single stream on real data.
Gap 4. Ground truth is noisy and models generalise poorly to new drivers	The project's model is trained on synthetic data and scores AUC = 1.000 on its own synthetic split, which is uninformative. The project therefore inherits this gap directly.	A small recorded, labelled dataset (KSS plus observer ratings (Wierwille & Ellsworth, 1994)) and subject-independent evaluation.

Gap	Project response so far	Still open
Gap 5. Detection is rarely evaluated together with graded, low-false-alarm intervention	Four graded alert levels with EMA smoothing, a 10 s persistence rule for the critical event, and escalating haptic, audio, visual and pause responses in one closed loop.	Measured time-to-alert, false alarms per hour in alert driving, and driver acceptance of the graded alerts.

Gap 1. Camera sensing is fragile, and fusion systems rarely cope with a missing modality

Evidence. Camera methods degrade with lighting, shadows, glasses, occlusion and head pose (Bai et al., 2022; Sikander & Anwar, 2019). Fusion studies usually sidestep the problem by excluding glasses-wearers (Daza et al., 2014) or discarding incomplete data (Schwarz et al., 2023). Feature-level fusion is acknowledged to fail when one signal drops (Samiee et al., 2014), and only recent work addresses it explicitly (Aravinth et al., 2025).

How the project responds. The vision pipeline now reports camera and face status, and the telemetry stream falls back to zeros after 5 s. In testing the face was detected in only 0 to 10% of frames under backlighting, so the gap is real for this system.

What remains open. Graceful degradation to a driving-only score when the face is lost; per-driver calibration of EAR and MAR thresholds; measured robustness across lighting conditions.

Gap 2. Vehicle-based measures are confounded by road context and plateau on their own

Evidence. Road context causes false alarms (McDonald et al., 2018; Li et al., 2017), studies often restrict analysis to straight roads (Forsman et al., 2013), driving-only indicators plateau around 0.72 to 0.74 (Sandberg et al., 2011), and the field relies on simple averaged metrics (Liu et al., 2009). Reaction-time effects are scenario dependent (Mahajan & Velaga, 2020).

How the project responds. Steering instability, reversals and corrections are computed on steering error relative to the curve, a wind-gust probe measures reaction delay, and driving features are fused with ocular features rather than used alone.

What remains open. Validation of the curve-normalised features against human drivers, not only autopilot profiles; testing on varied road geometry.

Gap 3. Fusion research relies on intrusive sensors and offline, simulator-only evaluation

Evidence. Physiological detection reaches about 90% but offline (Borghini et al., 2014); many fusion systems use EEG, ECG or EOG electrodes (Peng et al., 2024; Martensson et al., 2019). Reviews call for hybrid, non-intrusive systems (Sahayadhas et al., 2012), and evidence suggests non-contact camera plus vehicle sensing keeps most of the accuracy (Gwak et al., 2020; Daza et al., 2014).

How the project responds. A low-cost, contact-free loop: webcam plus simulator, feature-level fusion of exactly the variables the literature ranks highest (PERCLOS, SDLP, steering reversals) (Wang & Xu, 2016), scored live at 1 Hz on a laptop CPU.

What remains open. Measured end-to-end latency (rarely reported; one embedded system managed about 14 fps (Florez et al., 2024)) and an ablation showing whether fusion beats each single stream on real data.

Gap 4. Ground truth is noisy and models generalise poorly to new drivers

Evidence. KSS shifts with context and relates curvilinearly to physiology (Åkerstedt et al., 2014; Åkerstedt & Gillberg, 1990); its link to objective lapses is moderate (Kaida et al., 2006). Models lose much of their accuracy on unseen drivers (Martensson et al., 2019; Bakker et al., 2022; Maior et al., 2020), and simulator data may not transfer to roads (Hallvig et al., 2013; Wynne et al., 2019). A recent review names subjective labels and the lack of on-road data as the main barriers to deployment (Perkins et al., 2023). No reviewed journal paper trains or validates a drowsiness model on synthetic data.

How the project responds. The project's model is trained on synthetic data and scores AUC = 1.000 on its own synthetic split, which is uninformative. The project therefore inherits this gap directly.

What remains open. A small recorded, labelled dataset (KSS plus observer ratings (Wierwille & Ellsworth, 1994)) and subject-independent evaluation.

Gap 5. Detection is rarely evaluated together with graded, low-false-alarm intervention

Evidence. Detection papers report accuracy but seldom what the driver experiences. False alarms remain substantial (1.7 per hour (Fujiwara et al., 2019); 29% of awake periods (Li et al., 2017)), timeliness depends on how drowsiness is labelled (Schwarz et al., 2023), and most systems output a binary label (Gwak et al., 2020). Multistage alerts outperform single alerts but were tested only on short drives (Gaspar et al., 2017), false alarms erode compliance (Naujoks et al., 2016), and vibration can annoy drivers (Petermeijer et al., 2016).

How the project responds. Four graded alert levels with EMA smoothing, a 10 s persistence rule for the critical event, and escalating haptic, audio, visual and pause responses in one closed loop.

What remains open. Measured time-to-alert, false alarms per hour in alert driving, and driver acceptance of the graded alerts.

6. Aim and objectives

Aim

To design, build and evaluate a low-cost, contact-free, real-time driver fatigue detection system that fuses webcam facial cues with driving behaviour and issues graded alerts, and to quantify how well it works on real drivers.

The objectives are scoped for a small team over roughly one semester: each uses equipment already in the project (a laptop webcam and the browser simulator), a modest participant study, and success measures taken from the PRD or the reviewed literature.

Objective	Gap	Success measure	Status in repo
O1. Build and characterise a robust webcam pipeline with face-tracking status.	Gap 1	Face tracked in at least 90% of frames under frontal light; status reported within 1.5 s; documented failure conditions.	Pipeline and status reporting done; calibration and fallback to do.
O2. Develop curve-normalised driving features and show they separate alert from drowsy driving.	Gap 2	Each feature differs between conditions with a large effect size (Cohen's d above 0.8) on human data.	Features and autopilot profiles done; human comparison to do.
O3. Fuse both streams at feature level and test whether fusion adds value.	Gap 3	Fused AUC at least 0.87 and F1 at least 0.82 (PRD targets), fused AUC above the best single stream, inference under 100 ms, detection latency under 4 s.	Fusion pipeline done; ablation and latency measurement to do.
O4. Replace synthetic training data with a small labelled dataset.	Gap 4	Dataset collected with ethics approval; model reported on unseen drivers with confusion matrix and AUC.	Not started; data logging hooks exist (sessions.csv).
O5. Evaluate the graded alert loop as the driver experiences it.	Gap 5	Median time-to-alert under 30 s after observer-rated drowsiness; fewer than 2 false high-or-critical alerts per hour of rested driving; acceptance reported.	Alert levels, smoothing and persistence done; evaluation to do.

How each objective will be met

O1. Build and characterise a robust webcam pipeline with face-tracking status. Record tracking rate and feature stability for each participant under frontal, dim and backlit lighting, with and without glasses; add per-driver EAR baseline calibration in the first 60 s; fall back to a driving-only score when no face is seen for over 2 s.

O2. Develop curve-normalised driving features and show they separate alert from drowsy driving. Compare the six telemetry features between alert and drowsy runs (autopilot profiles first, then participants in rested vs sleep-restricted sessions).

O3. Fuse both streams at feature level and test whether fusion adds value. Train LightGBM on vision-only, driving-only and fused feature sets; leave-one-subject-out cross-validation; measure inference and end-to-end latency.

O4. Replace synthetic training data with a small labelled dataset. 10 to 12 participants, two 30-minute simulator sessions each (rested and after restricted sleep or late evening), KSS every 5 minutes plus observer ratings of face video; retrain and report subject-independent results.

O5. Evaluate the graded alert loop as the driver experiences it. During O4 sessions log time from first observer-rated drowsiness to each alert level, false alarms during rested driving, and a

short acceptance questionnaire.

7. Demonstrating understanding of the domain and key concepts

This section is a study guide. To show understanding you should be able to explain the domain in plain words, derive each feature, connect every design choice to evidence, and state the system's limits before anyone asks.

7.1 Domain essentials

Fatigue, drowsiness, sleepiness. Sleepiness (drowsiness) is the physiological pressure to fall asleep, driven by sleep loss and the circadian low between roughly 02:00 and 06:00 and after lunch. Fatigue is broader: reduced capacity from time-on-task or monotony, which can occur without sleepiness. Most detection systems, including this one, measure the observable signs of sleepiness. Say this distinction clearly; examiners often test it.

Microsleep. A brief lapse (about 0.5 to 15 s) with eye closure and no response. At 100 km/h a 4 s microsleep covers over 100 m. This is why eye-closure duration and reaction-delay features matter.

Why non-intrusive and multimodal. Physiological sensors (EEG, ECG) are accurate but need electrodes. Camera and vehicle signals are contact-free but each fails in its own way: cameras lose the face in poor light or when the head turns, and lane or steering signals are confounded by road curvature and driving style. Combining them is meant to cover each modality's blind spots.

Ground truth. There is no direct measurement of sleepiness. Studies label data with self-report (Karolinska Sleepiness Scale, 1 to 9), trained observer ratings of video, EEG markers, or experimental conditions such as sleep deprivation. Every label choice brings noise, and this limits every model trained on it.

7.2 Key concepts, formulas and where they live in the code

Eye Aspect Ratio (EAR)

Ratio of the eye's vertical openings to its width, computed from six landmarks per eye and averaged over both eyes. It drops towards zero when the eye closes and is roughly invariant to face size and distance.

$$EAR = (|p2 - p6| + |p3 - p5|) / (2 |p1 - p4|)$$

In the code: `backend/vision/features.py` (`eye_aspect_ratio`); landmark indices in `backend/vision/main.py` (left eye 33, 160, 158, 133, 153, 144).

Blink detection with hysteresis

A blink is counted when EAR falls below 0.23 (closed) and later rises above 0.26 (open). Two thresholds stop noise around a single threshold from counting as many blinks.

In the code: `backend/vision/main.py`; thresholds in `backend/vision/features.py`.

PERCLOS

Classically P80: the proportion of time the eyelids are at least 80% closed over a window (often 1 to 3 minutes). The project uses a proxy: the fraction of frames in the 30 s window with EAR below 0.23. Be ready to explain that difference.

$$PERCLOS = \text{frames with } EAR < 0.23 / \text{total frames in window}$$

In the code: backend/vision/feature_extractor.py.

Eye-closure duration (ECD_max) and blink frequency

ECD_max is the longest continuous run of closed-eye frames in the window, in seconds; long closures signal microsleep. Blink frequency is blinks per minute over the window.

$$BF = \text{blinks in window} / \text{window duration (s)} \times 60$$

In the code: backend/vision/feature_extractor.py.

Mouth Aspect Ratio (MAR) and yawns

Three vertical lip distances over twice the mouth width. A yawn starts when MAR exceeds 0.6 and ends when it falls below 0.5 (hysteresis again).

$$MAR = (|p_{13} - p_{14}| + |p_{82} - p_{87}| + |p_{312} - p_{317}|) / (2 |p_{61} - p_{291}|)$$

In the code: backend/vision/features.py (mouth_aspect_ratio).

Head pose (pitch)

Six 2D landmarks are matched to a generic 3D face model with OpenCV solvePnP. The rotation vector is converted to a matrix (Rodrigues) and decomposed into Euler angles. Pitch captures nodding. Because the model is y-up and the image y-down, the raw pitch sits near $\pm 180^\circ$; the code wraps it into $\pm 90^\circ$ so a level head reads about 0° .

In the code: backend/vision/headpose.py.

Sliding window aggregation

Frame-level signals are noisy, so features are statistics over the last 30 s, recomputed every second (1 s step). Window length trades responsiveness against stability; it also bounds detection latency.

In the code: backend/vision/buffer.py (RollingFeatureBuffer) and simulator/index.html (WINDOW_S = 30).

Lane-keeping features

Lane offset mean is the mean absolute distance from the lane centre; lane drift variance is the variance of lateral position, the simulator's version of SDLP (standard deviation of lateral position), the classic vehicle-based sleepiness measure.

$$\text{lane_drift_var} = \text{Var}(\text{offset}), \text{lane_offset_mean} = \text{mean}(|\text{offset}|)$$

In the code: simulator/index.html (computeFeatures).

Steering features, normalised for curves

Steering instability is the standard deviation of steering error, meaning input minus the steering the curve requires. Reversals and corrections are counted on that error with a deadband, so following the road does not count; only back-and-forth over-correction does. This directly addresses the road-geometry confound reported in the literature.

$$\text{steering_instability} = SD(\text{steer} - \text{required_steer})$$

In the code: `simulator/index.html`.

Reaction delay

Random wind gusts push the car sideways. Reaction delay is the time from gust onset until steering moves against the push, minus a 0.25 s keyboard and display baseline, averaged over gusts in the window.

In the code: `simulator/index.html` (`recordReaction`).

Context features

Session duration captures time-on-task. Time of day is encoded as sine and cosine so 23:00 and 01:00 are close together.

$$\text{tod_sin} = \sin(2\pi h / 24), \text{tod_cos} = \cos(2\pi h / 24)$$

In the code: `backend/ml/features.py`.

Feature-level fusion

Fusion can happen at the data, feature or decision level. Here features from both streams are concatenated into one 19-dimensional vector in a fixed order (`backend/ml/feature_schema.py`) and one model learns their interactions. Decision-level fusion would instead combine the outputs of separate per-modality models.

In the code: `backend/ml/features.py` (`build_feature_vector`).

LightGBM

A gradient-boosted decision tree library: trees are added one at a time, each fitted to the errors of the ensemble so far. LightGBM grows trees leaf-wise and bins features into histograms, which makes it fast on tabular data. Settings here: 200 trees, learning rate 0.05, max depth 6, 31 leaves, 80% row and column subsampling.

In the code: `backend/ml/train_model.py`; model file `backend/ml/model/fatigue_model.pkl`.

Exponential moving average (EMA)

Smooths the per-second probability so single noisy frames do not trigger alerts. With $\alpha = 0.3$ each new value carries 30% weight; a sudden change is about 66% absorbed after 3 s and 97% after 10 s.

$$S(t) = \alpha \cdot x(t) + (1 - \alpha) \cdot S(t - 1), \alpha = 0.3$$

In the code: `backend/ml/smoothing.py`.

Graded alert levels and persistence

Normal below 0.30, mild 0.30 to 0.55, high 0.55 to 0.75, critical above 0.75. The emergency-contact flag requires the score to stay above 0.75 for 10 consecutive seconds, a persistence rule that trades latency for fewer false alarms.

In the code: `backend/session_manager.py`; simulator alerts in `simulator/index.html`.

Evaluation metrics

ROC-AUC measures ranking quality across all thresholds. Precision is the share of alarms that were real; recall is the share of drowsy periods caught; F1 is their harmonic mean. Report a confusion matrix, and evaluate subject-independently (leave-one-subject-out), because random splits leak each driver's personal baseline into the test set.

$$F1 = 2 \cdot \text{Precision} \cdot \text{Recall} / (\text{Precision} + \text{Recall})$$

In the code: `backend/ml/train_model.py` prints accuracy, AUC, confusion matrix and report.

7.3 Limitations you should state yourself

- **Synthetic training data.** The 5,000-row training set (balanced 2,500 / 2,500) is generated, not recorded. On its own held-out split the model scores AUC = 1.000 and F1 = 1.000 because the synthetic classes separate perfectly. This proves the pipeline runs end to end; it says nothing about real-world accuracy. Say this before an examiner does.
- **Demo bias layer.** On top of the model, each detected yawn adds 0.15 and each blink 0.01 to the score, decaying by 0.02 per second (`backend/pipeline_interface.py`). It is a hand-tuned heuristic for live demos, not a learned effect.
- **PERCLOS proxy.** The code's PERCLOS is the share of frames below an EAR threshold over 30 s, not the classic P80 measure over minutes.
- **Fixed thresholds.** EAR and MAR thresholds are the same for every driver, although baseline eye openness varies between people and with glasses.
- **Lighting and pose.** In testing, the face was detected in 0 to 10% of frames when the room was backlit. The system now reports 'No face found', but the score cannot update without a face.
- **Simulator validity.** The browser simulator is a top-down game with keyboard steering. Its telemetry features were calibrated to the ranges of the training data, not validated against real driving.
- **Emergency contact.** The critical event is logged; no SMS or call is sent yet.

7.4 Likely questions and model answers

Why combine camera and driving data instead of using one?

Each fails differently. The camera loses the face in poor light or with head turns; lane and steering measures are confounded by curves and driving style and tend to change late. Fusion lets one stream compensate for the other, and the reviewed fusion studies generally report higher accuracy than single-modality baselines.

What does EAR measure and why use a ratio?

Eye openness: vertical eyelid distances over eye width. Because it is a ratio, it barely changes with distance to the camera or face size, so one threshold works across frames.

Why two thresholds for blinks and yawns?

Hysteresis. With one threshold, a signal hovering near it would count many events. Requiring a drop below 0.23 and a rise above 0.26 counts one clean blink.

How is PERCLOS defined, and how does your version differ?

Classic P80 is the share of time the eyes are at least 80% closed, usually over 1 to 3 minutes. Ours is the share of frames with EAR below 0.23 over 30 s: a faster proxy that suits a 1 Hz system, but not identical.

Why LightGBM rather than a deep network?

The input is 19 tabular features, where gradient-boosted trees are strong, fast (well under 100 ms per prediction on a CPU) and interpretable through feature importance. A deep temporal model would need far more labelled data than exists for this project.

Your model reports AUC = 1.0. Is it perfect?

No. That is on synthetic data whose classes are separable by construction. Real performance is unknown until the model is trained and tested on recorded, labelled sessions with subject-independent cross-validation (Objective 4).

How do you get ground truth for drowsiness?

Options are self-report (KSS), trained observer ratings of video, EEG markers, or controlled conditions like sleep restriction. The plan is KSS every few minutes plus observer ratings, accepting that both are noisy.

What does the EMA do, and what is the cost?

It suppresses one-second spikes so alerts do not flicker. The cost is lag: with $\alpha = 0.3$ a sudden change takes about 3 s to be two-thirds reflected, which must fit inside the detection-latency target.

How do you stop road curves from looking like drowsy steering?

Steering features are computed on steering error: input minus the steering the curve requires. Following a bend produces no error, so only over-correction counts.

What happens when the face is not visible?

The vision pipeline reports 'no face' and the simulator shows it. The model is only run when vision features exist, so the score holds its last value. Graceful degradation to driving-only scoring is future work tied to Gap 1.

Why four alert levels instead of one alarm?

Graded warnings escalate with risk: mild prompts, high warnings, critical intervention. The alerting literature links unnecessary or over-intense alarms to annoyance and loss of trust, so lower levels are kept gentle.

What would you change for a real vehicle?

Replace simulator telemetry with CAN-bus or OBD-II steering and lane-camera data, use an infrared camera for night and sunglasses, calibrate thresholds per driver, and validate on recorded drives with proper labels.

7.5 Checklist for demonstrating understanding

1. **Explain the domain in two minutes:** sleepiness vs fatigue, microsleep, the four measure families and their trade-offs (the table in Section 4), and why the project chose non-intrusive fusion.

2. **Derive every feature:** write the EAR, MAR, PERCLOS, EMA and F1 formulas from memory and point to the file that computes each one.
3. **Justify each design choice with a citation:** for example, PERCLOS and eye closure from the vision theme, SDLP and reversals from the driving theme, LightGBM from the ML theme, graded alerts from the alerting theme.
4. **Run the live demo in a fixed order:** start the server; open the simulator; wait for 'Tracking face'; drive alert, then switch to the drowsy autopilot (key 3) and show the telemetry panel change; yawn or close your eyes and show the score rise to a warning and the critical pause.
5. **Show evidence, not claims:** the confusion matrix, an ablation (vision-only vs driving-only vs fused) once real data exists, latency measurements, and the camera tracking rate under different lighting.
6. **Own the limitations** in Section 7.3 before you are asked, and link each to the objective that addresses it.
7. **Connect gaps to objectives to evidence:** be able to say which gap each feature or experiment answers and how you would know whether it worked.

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APA 7th edition. All 50 entries are journal articles verified against Crossref; DOIs link to the publisher record.

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